

Project Overview

A03 Neural Network Zoo – Project Overview

Our group chose the Convolutional Neural Network (CNN) and the chameleon as our Neural Network Zoo animal.

We chose the chameleon because its scanning behavior gives us a useful way to introduce how a CNN examines different parts of an image for patterns and features.

One idea that helped us understand CNNs was realizing that a computer does not first see a picture the way we do. It receives organized numerical data. Filters examine small regions of that data, detect useful patterns, and deeper layers combine those patterns into more meaningful features that can support a final prediction.

During our discussion, we first thought of the chameleon's two eyes as hidden layers. We realized the better analogy is that the eyes represent gathering visual information, while hidden layers represent stages of processing that information.

Our presentation will follow this path:

Image Data → Local Patterns → Features → Deeper Processing → Prediction

The goal is to explain how a CNN works from first principles, why it is useful for visual tasks, and how the chameleon helps make that process easier to understand.

Project Plan

- James Delgado: Neural network basics
 - Artificial neuron
 - Inputs, weights, bias, activation
 - How neurons form layers
 - Short deep learning intro
- King Owusu: CNN structure and how it works
 - What Convolutional Neural Network means
 - Filters/kernels
 - Feature maps
 - Hidden layers
 - How features build from simple to more complex
- Esther Moseka: Chameleon analogy and visuals
 - Chameleon scanning behavior
 - Connect that behavior to examining different image regions
 - Create the main diagram/visual
 - Explain where the analogy works and where it stops
- Person 4: Applications, limitations, and reflection
 - Real-world CNN uses
 - Why CNNs are useful for images
 - Limitations
 - Final reflection on what the group learned

Then everyone contributes their own 1–2 slides, and the group does one final review together.

The story stays:

Neuron → Network → CNN → Chameleon → Applications → Reflection

Project Journal

James Delgado Journal Entry:

This project helped me understand Convolutional Neural Networks (CNNs) in a more practical way instead of just memorizing terms. One of the biggest things that helped was connecting the artificial neuron back to what I already knew from biology. A biological neuron receives signals, combines them, and passes information forward. An artificial neuron is much simpler, but the general idea of receiving inputs, weighting them, and passing an output helped make the network structure easier to understand.

I also spent more time thinking about why it is called a Convolutional Neural Network instead of just accepting the name. What made it click for me was the idea of a small filter moving across different regions of an image and asking the same basic question in each location: is this pattern present here? The sliding reveal toy was a useful comparison because it gave me a physical way to picture a moving window, even though the CNN is calculating rather than simply revealing part of an image.

The chameleon ended up being a good animal for our project because its scanning behavior connected naturally to visual processing. At first, I thought its two eyes might represent hidden layers, but that comparison was not quite right. The eyes are better used as an analogy for gathering visual information, while hidden layers are stages of processing. Correcting that idea actually helped me understand the architecture better.

Overall, this project helped me see CNNs as a sequence of simple ideas working together rather than one complicated AI concept.

King Owusu—CNN Structure and How It Works.

A Convolutional Neural Network (CNN) is a type of artificial neural network that is mainly used to work with visual information, especially images. Instead of looking at an entire image all at once, a CNN examines smaller areas and learns useful patterns from them. A CNN is made up of an input image, convolutional layers, filters or kernels, feature maps, hidden layers, and an output layer. These parts work together to take the original pixels of an image and turn them into information that can be used to recognize or classify what is in the image.

One of the main parts of a CNN is the filter, also called a kernel. A filter is a small matrix of numbers that moves across different parts of an image. As it moves, it performs calculations on the pixels in each area to look for specific patterns. Different filters can learn to detect things such as edges, lines, colors,

corners, and textures. The information produced by applying a filter to an image is called a feature map. It shows where a certain feature has been found in the image and gives the CNN useful information to work with.

CNNs have several hidden layers that help the network learn features at different levels. The first layers usually focus on simple features, such as edges and lines. As the information moves through deeper layers, these simple features are combined into more detailed patterns. For example, the network may first recognize lines and curves, then shapes and textures, and later parts of an object. The deeper layers can combine these different features to recognize the whole object. This allows the CNN to learn what is important in an image without having to manually identify every feature.

For our project, the chameleon is a good way to represent how a CNN examines an image. A chameleon can look at different parts of its surroundings, while a CNN uses filters to examine different regions of an image. The network does not recognize an object immediately. Instead, it gradually builds an understanding by combining information from different layers. For example, when looking at an image of a chameleon, the CNN could start by detecting pixels and edges, then recognize shapes and textures, and eventually identify features that help it recognize the chameleon. Learning about this process helped me understand how CNNs break down complex images into smaller patterns and then combine those patterns to make a prediction.

Zain Ahmed – Applications, Limitations, and Reflection

For this assignment, my main contribution to the group was working on the real-world applications, limitations, and final reflection for CNNs. For the applications part, I looked at different ways CNNs are used with images in the real world. Some examples I included were medical imaging, self-driving cars, security cameras, and factory quality control. I also explained why CNNs are useful for images because they can look for patterns like edges, shapes, and textures and build those into more complex features.

I also worked on the limitations of CNNs. One thing I learned is that CNNs depend a lot on the data they are trained with. Things like lighting, image angle, blur, noise, or rotation can sometimes affect the prediction. I also talked about how a CNN does not actually understand an image the same way a person does. It is mainly recognizing patterns from the image data that it learned during training.

For the final reflection, I connected my part back to what everyone else in the group worked on. We started with artificial neurons, then explained neural networks and CNNs, and used the chameleon analogy to make the process easier to understand. Overall, working on my part helped me understand more about where CNNs can actually be used and also why we should not automatically trust every prediction a model makes.

Esther Odaibo

Understanding CNNs Through the Chameleon Analogy

For my part of the project, I created two slides explaining the chameleon analogy and its limits. A chameleon examines different areas of its surroundings with its eyes, while a CNN uses filters to examine small regions of image data for patterns such as edges, colors, and textures. Deeper layers combine these features to support a prediction.

I also learned that the chameleon's eyes should not be compared to hidden layers. The eyes represent gathering visual information, while hidden layers process and combine that information. This project showed me that an analogy can make CNNs easier to understand, but its limits must be explained so that it does not give a false impression of how the network works.